PALLOR

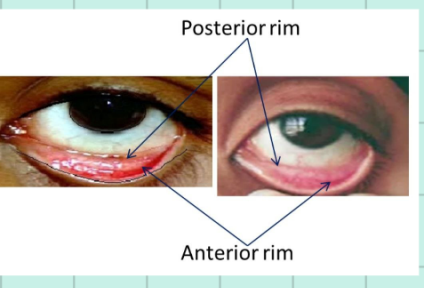
oxy Hb →

Areas of superficial vessels → oxy Hb → red colour → visible or not

⊖ of Pallor doesn't r/o anemia

Location →

1. Anterior rim of lower palpebral conjuctiva
2. Tongue/oral mucosa
3. Palm
4. Nail bed



Causes of pallor

Conc^{n} of Hb ↓(oxy) in these vessels

① Hb↓ → Anemia → ♂ → <13g%
→ ♀ → <12g%

Koilonychia → Fe def.

Knuckle hyperpigmentation → B12

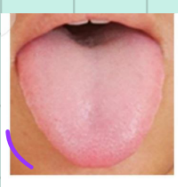
Bleeding tendencies → Bone marrow

When visible? Grading?

Skin & mucous memb → <10gm%

Crease less pigmented than surrounding → <8gm%

Hyperextended hand



Cond^{n}s in which pallor ⊕ but no Anemia

Blood flow ↓ → Shock / LHF / PVD

Blood pressure ↓

Addison

Hypopituitarism (cortisol → ↑ rate NE ↓ → SNS ↓ → BP ↓)

Hypothyroidism → (Tho → ↑ + Thy − ↓ → SNS ⊖)

Normal nail bed color

Pallor nail bed

Severe nail bed pallor

Icterus

Yellowish discoloration of eyes & skin

<2.5mg/dl → Latent

During reversal → can be seen at <2.5mg/dl d/t binding ↓ albumin

When >2.5mg/dl

Areas to look for

UPPER SCLERA

why?

elastin in conjuctiva + overlying the sclera *

HARD PALATE

UNDERSURFACE OF TONGUE

HANDS

① MUDDY LOWER SCLERA

② LIGHT EXPOSURE → convert to

ALWAYS VISUALISE IN DAY LIGHT

DDx

- **Jaundice** (↑ Bilirubin) + Sclera +/- Skin
- **Hypercarotenemia** (↑carotene) ⊖ Sclera ⊕ → mainly palm, sole, forehead & nasolabial fold
  - ↑ vit. A intake
  - Hypothyroidism

Raised if T. Bil is >0.8mg/dl

soluble

Prehepatic

↓

Lemon coloured

Hemorrhage collection

Chipmunk facies (eg thalassemia)*

Hepatic

↓

Greenish Yellow

Signs of liver failure

Posthepatic

↓

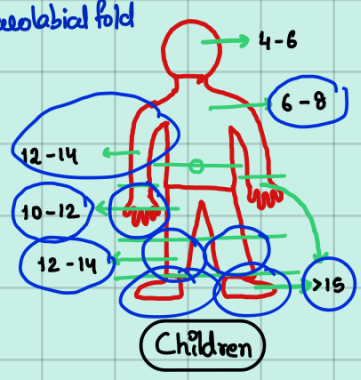
Dark Yellow

Characteristic scratch marks ± bleeding point

Bradycardia (d/t bile salts)

When visible? Grading?

Krammer zone



DON'T COMMENT ON PALLOR IF ICTERUS PRESENT

CYANOSIS

Blue | Red

↓

Bluish discoloration of the areas of body ∵ superficial vessels

LOCATION

① Nose tip	① Ear lobes
② Lips	② Finger tips
③ Buccal mucosa	
④ Tongue	

CAUSES

Pseudo ↑ Metals / Drugs (Au, Ag) (Amio, Mino)

- deoxy Hb → >5g%
- meth Hb → >1.5g%
  - congenital
  - drugs / toxins → NTG, Dapsone, LA
- sulf Hb → >0.5g%
  - drugs / toxins → sulfonamides

Central cyanosis

↓ O_2 in air → High altitude

↓ O_2 entering blood
- OSA
- Epiglottitis

(upper)

- Severe acute exacerbation of asthma / COPD (lower)
- Pneumonia, ARDS (alveoli)

↓ blood flow / abN blood flow b/w heart & pulmonary & aortic vessels → Cyanotic Heart D/os
- → CHF

Peripheral cyanosis

Local BVs (of a region) →
- Obstruction → PAD / Emboli / HVS
- Vasospasm → Raynaud's

↓ O_2 to tissues → take out more from the Hb → ↑ deoxy Hb

CHF

D/o ∵ both central & peripheral → CHF

Pt. comes ∵ bluish discoloration → DDx → ↑ deoxy, ↑ meth, ↑ sulph → DDx → Central / Peripheral

Location

- Lips
- Buccal mucosa
- Nose tip
- Ear lip
- Finger tip
- Circumoral

Extremities

- Warm
- Cool

↓ on warming of extremities

- No
- Yes

Clubbing & Polycythemia

- Usually associated
- Not associated

Response to 100% oxygen

- Cyanosis ↓ (>5g%)
- Cyanosis persist

(IT IS ABSOLUTE RATHER THAN RELATIVE QUANTITY REQUIRED)

Cyanosis in Severe anemia → less likely as Hb% is low, >5g% is difficult to reach.

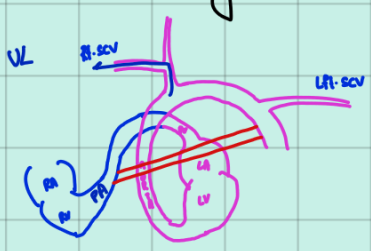
Cyanosis in Polycythemia → More chances as ↑ Hb%.

Reddish tongue d/t Polycythemia + Bluish d/t cyanosis

↓

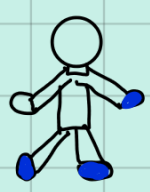
Ruddy cyanosis

Differential cyanosis → Upper limbs (N), lower extremities cyanosed → PDA ± eisenmengerisation

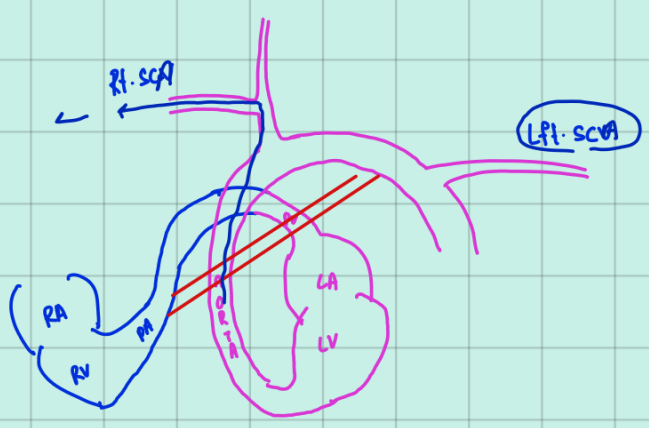


Reverse differential cyanosis → UL cyanosed, LL (N) → PDA ± TGA

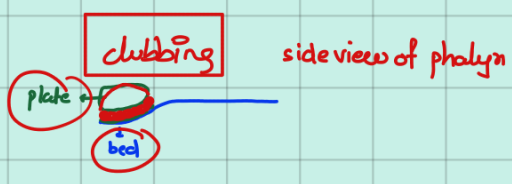
3/4 cyanosis →



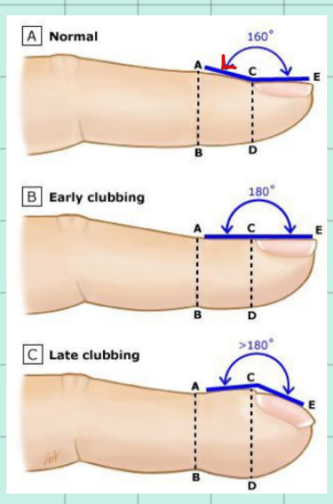
PDA proximal to left SCA



Clubbing



Bulbous (like a bulb) enlargement of the distal phalanges of fingers & toes d/t proliferation of the connective tissue beneath the nail plate

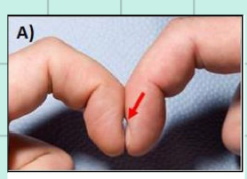


① HOW TO WE KNOW IF CLUBBING IS ⊕ or NOT?

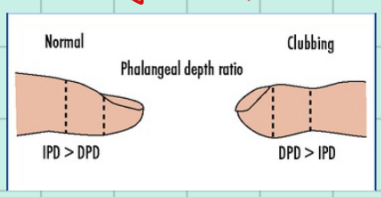
① Fluctuation of nail bed

② Lovibond L

③ Shamroth window



④ Phalangeal Depth ratio

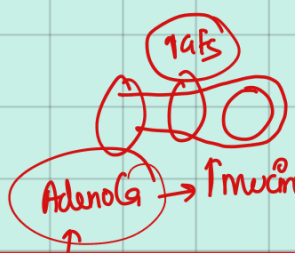


② CAUSES & MECHANISMS

THEORIES

- HYPOXIA occurs → ↑ release of GFs (VEGF, V, C)
  - Respiratory cond'n: Abscess / Bronchiectasis / Cystic fibrosis / Empyema / lung Cancer & Mesothelioma / ILD
    - x COPD, x TB itself
    - If ⊕ → lung Ca
    - If ⊕ → d/t 2° fibrosis / Bxis
  - Cyanotic HD / Acyanotic HD → Eissenmengenisation
- Platelet comes to nail bed bringing PDGF → them
  - Emboli → Infective Endocarditis
- Chronic inflammation → ↑ GFs / vasodilators
  - Hepatic Cirrhosis ✓
  - IBD ✓
  - Celiac Disease ✓
- Neurogenic → ↑ PSNS → ↑ PSNS ⊕ → ↑ vasculdrity → ↑ blood → ↑ factors
  - ↓ PGs
- Congenital → inhibition of Prostaglandin Dehy → ↑ PGs → ↑ vasodilation
  - Pachyderma periostosis
  - Lionine facies
    - Hyperkeratosis of palms & soles
    - Clubbing

③ Grading



Thrombos-accn → Platelets → PDGF

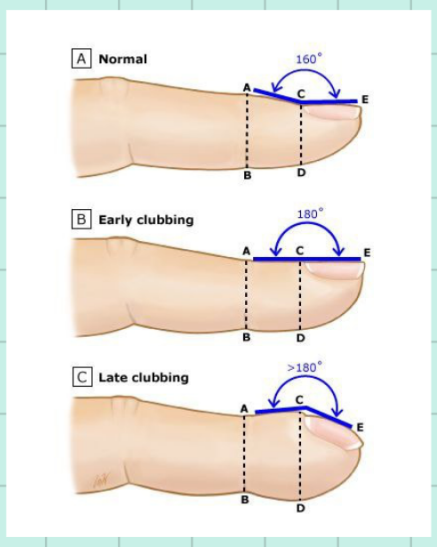
I → Fluctuation (+)

II → Loviband ∠ ↑↑ (>180°), Shamroth window obliteration

III → Parrot Beak / Drum stick app.

IV → Hypertrophic Osteoarthrophy (+) → DISTAL EPIPHYSES OF LONG BONES

★ Bronchogenic Ca as paraneoplastic syndrome



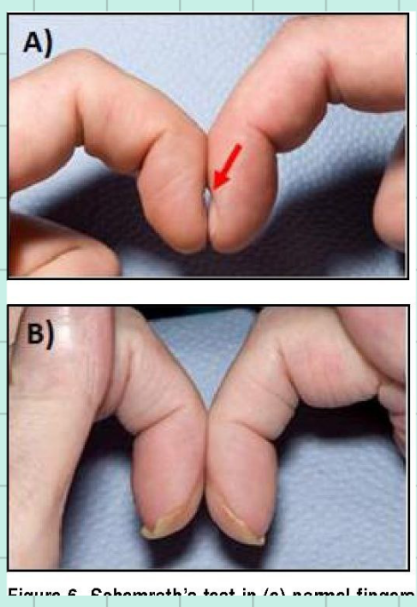


Figure 6 Schamroth's test in (a) normal fingers

Differential clubbing → UL (N)
LL toes clubbing (+) → PDA ± Eisenmengerisation

Reverse Differential Clubbing → UL clubbing (+) → " + TGA
LL (N)

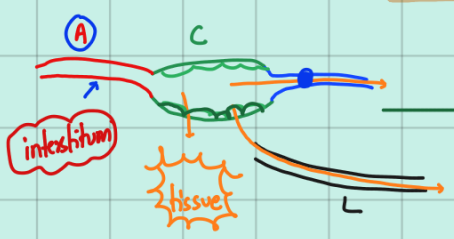
Unilateral clubbing → ① Cervical rib
② Subclavian artery aneurysm
③ Hemiplegia

Unidigital clubbing → ① Median N. palsy
② Trauma

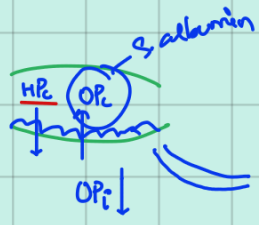
Edema → accumulation of fluid in extracellular space i.e. the interstitium

Anasarca → gross, generalised edema.

① MECHANISMS / CAUSES



Starling forces
Endothelial junct°
Lymphatic Drg.



mainly fluid acc° occur → Pitting

on pressing for 30s → depression created → resolving in
- <40s → D/t low oncotic pressure
- >40s → D/t high hydrostatic pressure

ⓐ ↑HPc → venous outflow obstruction → DVT | Varicose Vein | Pelvic or Intra-abdominal mass

↑ blood through arteries 2° to CKD → ↑R of Na^+ & H_2O by DCT → ↓ blood to kidney

CHF | CP → B/L

ⓑ ↓OPc → Hypoproteinemia → CLD → B/L
- Nephrotic Syndrome
- Malnutrition
- PLE

ⓒ ↑ permeability → Cellulitis

ⓓ ↑OPi → ↑ protein acc° → Hypothyroidism → B/L

Hardness (+) → Non-Pitting

ⓔ ↓ lymphatic drainage → B/L
- Tumour
- Trauma
- Filariasis
- Congenital
- Radiation induced

↑ fluid acc° → Initially pitting
+ protein acc°
- Lymph (+) fibroblasts → ↑ sclerosis → Lipodermatosclerosis
- fluid reabs through small channels → later non-pitting

② LOCATION → initial site is where fluid can gravitate down

Dependent pos°
- Ambulatory → Ankle → 2cm above Medial malleoli
- Non-Amb° → Sacral edema

Swelling of LL

B/L

Pitting
- CHF → Pedal → Generalised (less in morning → ↑ throughout day)
- CLD → Ascites → Pedal → Generalised
- Nephrotic Syndrome → Periorbital (more in morning) → Face → Generalised
- Protein losing Enteropathy → Diarrhea (+)
- CKD → ↓urine output

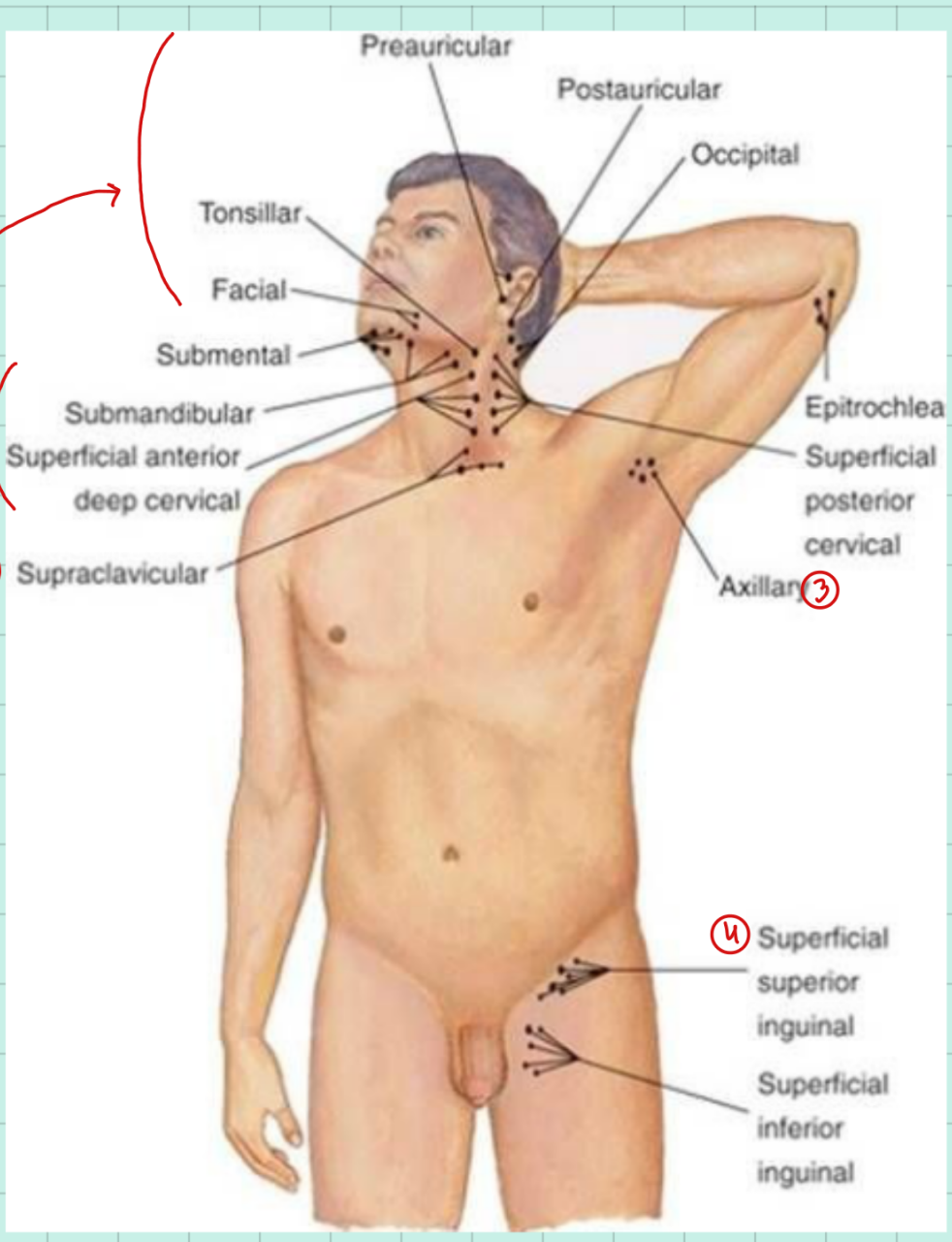
Non-Pitting
- Hypothyroidism
- B/L lymphedema → Stemmer sign (+)

U/L → Cellulitis → Pain/Redness
- Varicose Veins → Torturous veins
- DVT → severe bursting pain/immob°
- Lymphedema → Stemmer

DRUGS

Facial edema DD/
1. Nephrotic Syndrome
2. SVC obstruction, redness
3. Angioedema 2° to drugs/↓↓ ACE⊖ (dd - I)
4. Hypothyroidism
5. Trichinosis

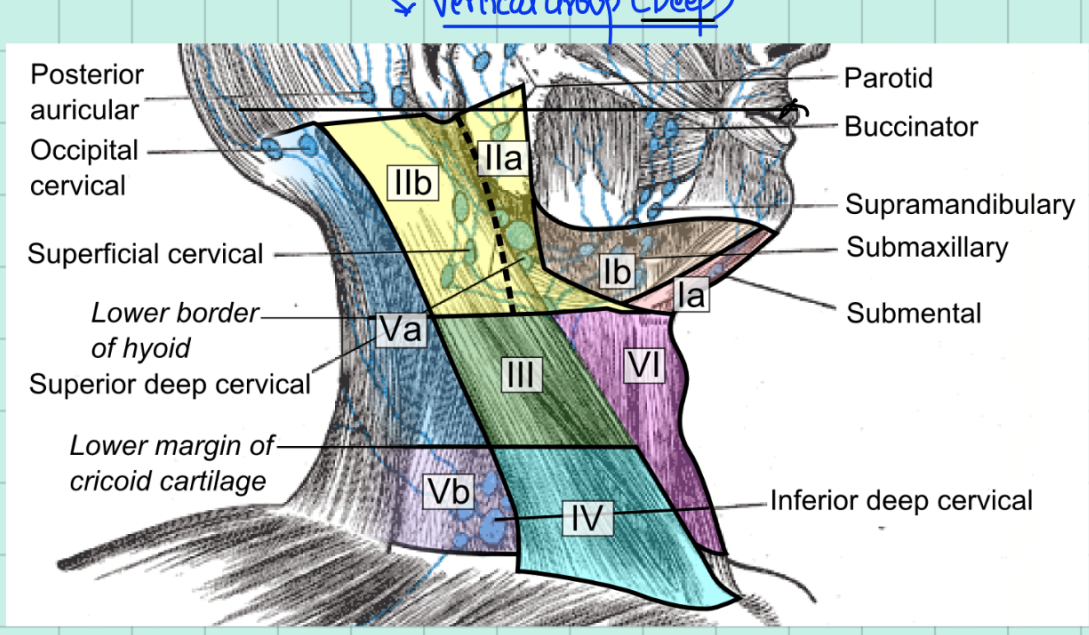
Drugs inducing edema
1. ACE⊖s
2. Diltiazem
3. Nitrates
4. Steroids
5. NSAIDs

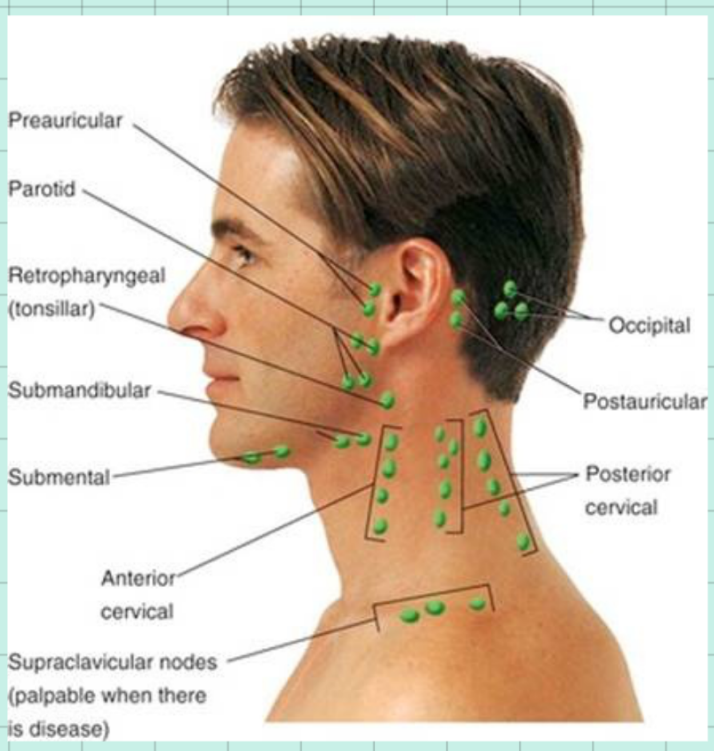


① Cervical LNs

① Cervical LNs → Horizontal Group (Superficial) → External Waldeyer Ring

① Cervical LNs → Vertical Group (Deep)





(b) Axillary LN

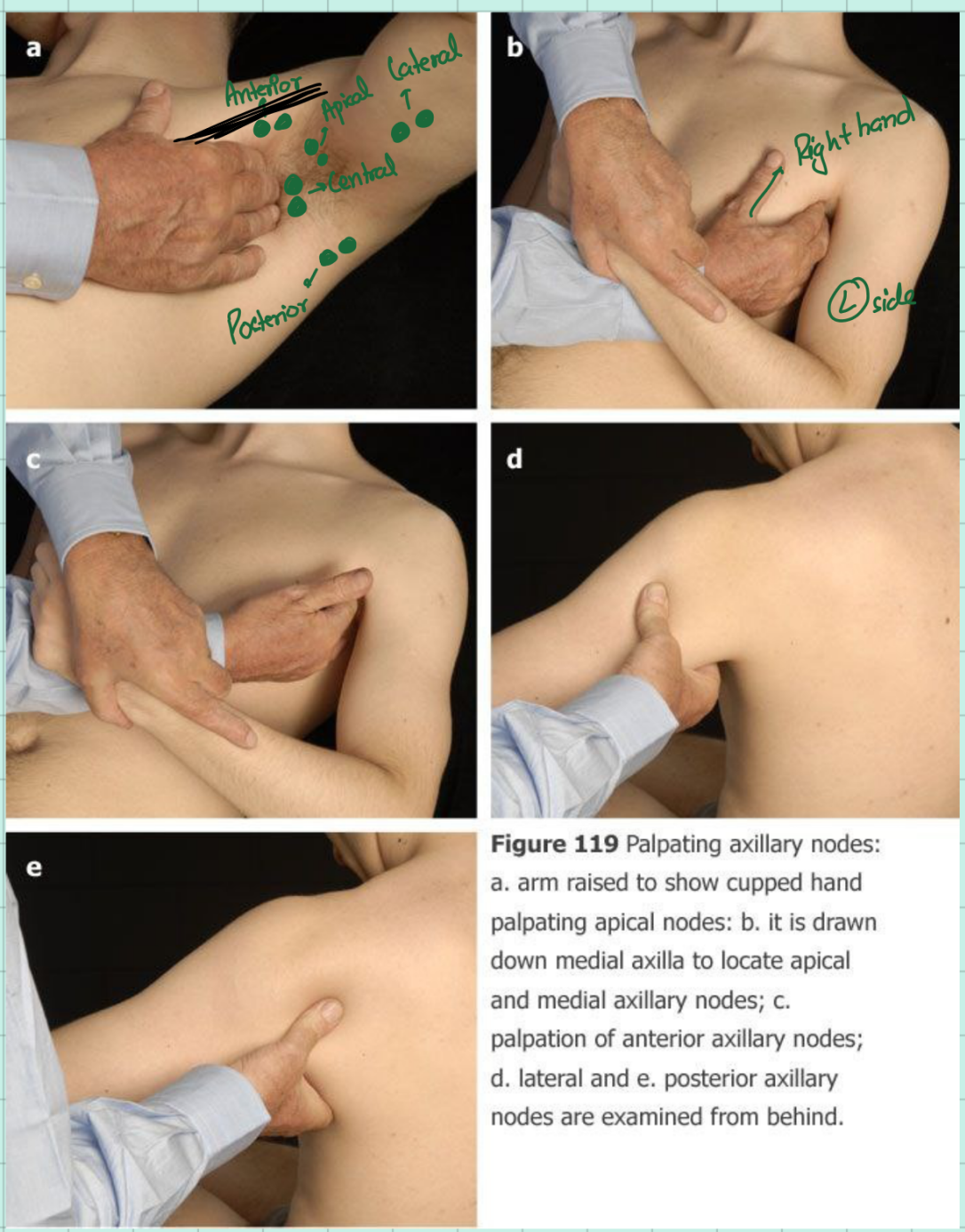


Figure 119 Palpating axillary nodes: a. arm raised to show cupped hand palpating apical nodes: b. it is drawn down medial axilla to locate apical and medial axillary nodes; c. palpation of anterior axillary nodes; d. lateral and e. posterior axillary nodes are examined from behind.

(c) Supraclavicular & Infraclavicular LN →

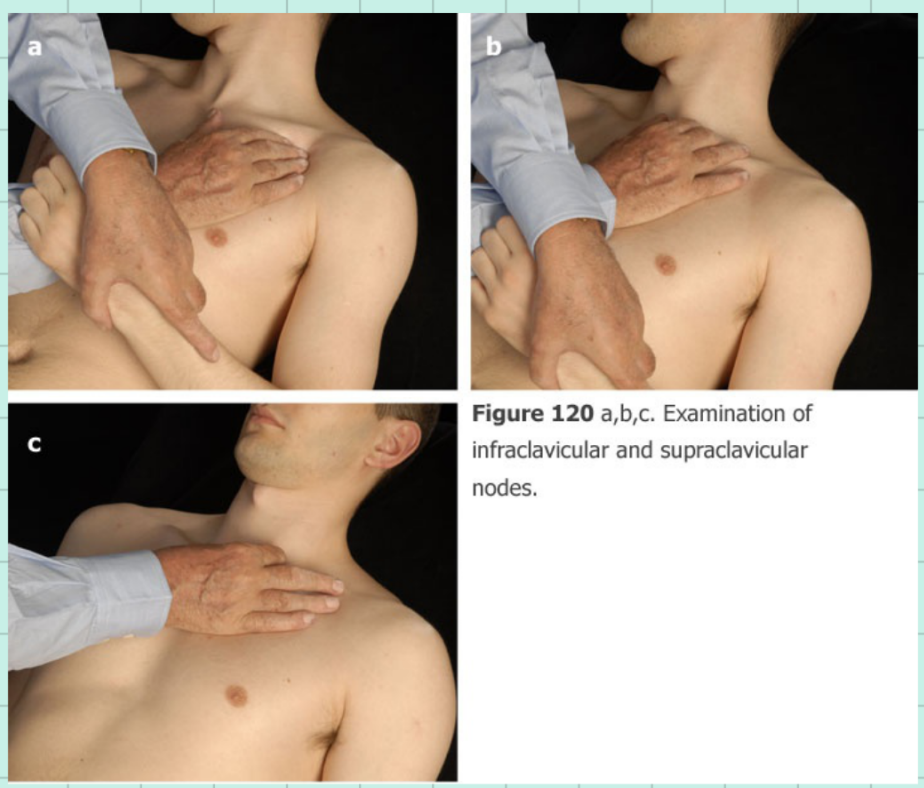
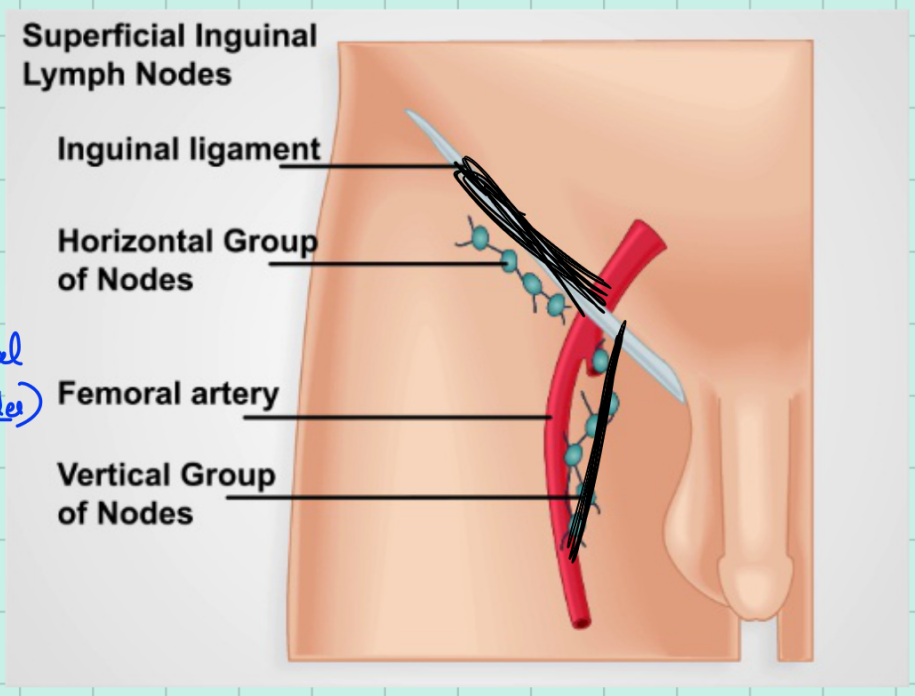


Figure 120 a,b,c. Examination of infraclavicular and supraclavicular nodes.

LEFT Supraclavicular LN ⇒ Virchow's node

(d) Inguinal LNs →

- Superficial
  - Horizontal
  - Vertical
- Deep → Femoral canal (Cloquet node)



Significant lymphadenopathy →

① Inguinal → >2cm

② Cervical / Axillary → >1cm

* Axillary LN → >1.5cm

③ Epitrochlear / Supraclavicular → Any size

Causes

Generalised or not?

→ >1 non-continous group of LNs involved

Y → Generalised

- If ⊕ for >3m → Persistent Generalised → *HIV
- Systemic diseases
  - Infections →
    - Bacterial → TB, Typhoid, Syphilis
    - Viral → EBV, HIV
    - Parasitic → Toxoplasmosis
    - Fungal → Histoplasmosis
  - AI d/Os → SLE / Sarcoidosis / Sjogren

N → early course

- Disease of local draining area
  - ① Cervical LNs →
    - Infections → Tonsillitis / Pharyngitis / Post. 1/3rd nose / TB → *** → JD LN → Wood LNs
    - Neoplasm → Tonsillar Ca / NPC / Laryngeal Ca / Thyroid Ca / Lung Ca

HS → Kawasaki

Neoplastic → Lymphomas
Leukemias

Drugs → Phenytoin
ACE Θrs → Captopril
NSAIDs
Carbemezapine

② Axillary LNs →

Infections → Limb infection / Breast abscess

Neoplasms → Breast Cancer
Malignancy on limb, eg. Melanoma.

③ Supraclavicular LN →

Enlargement of left → Troisier Sign
d/t Reflux theory

** Important causes →
Breast Malignancy
Lung Malignancy
GI malignancies
Bladder Ca
Gonadal Malignancy *

(*) left supraclavicular LNs should always be checked in P/A examination

④ Inguinal LNs

Horizontal
↓
Scrotum
Anus below Dentate Line
Perineum
External Genitalia
↓
into Para-aortic LNs because they come from abdomen.

presents as epigastric lump

Draining lower limb
↓

In case of high tumour load / tumour seeding into scrotal skin, Inguinal LNs are involved

Vertical
↓

⑤ Epitrochlear LNs → 5cm above trochlea, on medial aspect of arm.

* Syphilis
* Cat Scratch disease

HOW TO GUESS ETIOLOGY ON EXAMINATION -?

① Tenderness +/- → infectious condition

② Consistency →
- HARD → MALIGNANCY
- SOFT / FIRM → Ⓝ / INFECTIONS / AIDs
- RUBBERY → HODGKIN LYMPHOMA

(3) Discrete / Matted → TB → Periadenitis

(9) Mobility → used in malignancy
↳ may be used in chronic infections